BELLWETHER-2 — Preregistration

Series: BELLWETHER (spatial nonclassicality witness)
Experiment ID: BELLWETHER-2
Owner: Remnant Fieldworks Inc. (Derek Adam Hone)
Status at lock: design + harness complete, validated in simulation, hardware job pending
Governing framework: RF Experiment–Pilot–IP Covenant (preregistration-first; honest outcomes; claims narrower than evidence; file-first / publish-after for new matter)

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1. One-sentence statement

We measure the **3-qubit Mermin inequality** (genuine multipartite entanglement) on a GHZ state and **cryptographically bind the measured M statistic into an authorization ProofRecord** as a multipartite spatial nonclassicality witness — a certificate that three independent parties must jointly agree.

2. Why this is new (advancement over BELLWETHER-1)

Series	State	Test	Witnessed
BELLWETHER-1	2-qubit Bell+	CHSH (S>2)	**bipartite** entanglement (Alice + Bob)
BELLWETHER-2	3-qubit GHZ	Mermin-3 (|M|>2)	**genuine multipartite** entanglement (Alice + Bob + Charlie)

Key advancement: Mermin violations grow exponentially with # qubits and certify **genuine** (not just pairwise) multipartite entanglement. BELLWETHER-1 tested 2-party entanglement; BELLWETHER-2 tests 3-party joint nonclassicality.

Authorization angle: multipartite witness models **multi-authority authorization** — multiple independent parties must jointly agree. The ProofRecord binds physical evidence that the authorization value cannot be replicated by any pair acting alone.

3. Physics and design

State: 3-qubit GHZ state
$|GHZ\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$

Test: Mermin-Klyshko inequality for 3 qubits:

$$M = \langle A_1B_1C_1\rangle + \langle A_1B_2C_2\rangle + \langle A_2B_1C_2\rangle - \langle A_2B_2C_1\rangle$$

where A_1,A_2 are Alice's measurement choices (X,Y), B_1,B_2 are Bob's (Y,X), C_1,C_2 are Charlie's (Y,X).

Four measurement circuits (in operator notation):
- **XYY**: X⊗Y⊗Y
- **YXY**: Y⊗X⊗Y
- **YYX**: Y⊗Y⊗X
- **XXX**: X⊗X⊗X

Each 3-bit outcome maps to a parity: (-1)^(popcount). Expectation value = shot-average of parities.

Bounds:
- **Classical (local-realistic) bound:** $|M| \leq 2$
- **Quantum (GHZ) prediction:** $|M| = 4$

Simulation validation (pre-lock, noiseless AerSimulator, 200k shots/setting):
E = [−1.000, −1.000, −1.000, +1.000] → **M = −4.000**
(Sign is a basis/phase convention; the violation is $|M| = 4 >> 2$, confirming maximal quantum value.)

4. Execution plan

- **Backend:** IBM open-plan hardware (prefer `ibm_fez`, else min pending-jobs).
- **Shots:** 2000 per setting × 4 = **8000 shots total** (~a few seconds of QPU time; sized to remaining open-plan budget).
- **Mode:** Batch (open plan forbids Session). **Channel:** `ibm_quantum_platform`.
- **Circuits transpiled** to backend ISA (optimization_level=1) before submission.

5. Outcome criteria (fixed in advance — honest, symmetric)

Verdict	Condition
VIOLATION (genuine multipartite entanglement, under assumptions)	$\|M\| > 2$ and $\|M\| - 2 > 3\sigma$
INCONCLUSIVE / HOLD	$\|M\| > 2$ but not 3σ-significant
NO VIOLATION	$\|M\| \leq 2$

The measured hardware M will be reported **exactly as observed**, whichever bin it lands in. No re-runs to chase a preferred number.

6. Honesty boundary (non-negotiable — carried into every artifact)

- **Locality loophole OPEN:** Co-located qubits on a single chip can in principle communicate classically. A determined local-realist can attribute any $|M| > 2$ to such signaling.
- **Detection loophole OPEN:** Imperfect readout means not all events are detected. Fair-sampling assumption required.

- This is therefore a **Mermin violation UNDER STATED TRUST ASSUMPTIONS**, **not** loophole-free, and **not** device-independent certified randomness.
- **Single device**, no device-independence is claimed.
- **Only defensible claim:** “Observed 3-qubit Mermin inequality violation (genuine multipartite entanglement) on quantum hardware, under locality + detection trust assumptions, cryptographically bound into the authorization record.”

6a. Disclosed pre-execution harness facts

- The proven WITNESS/BELLWETHER-1/CHRONO harness fixes are baked in: FIX-1 (robust backend selection via min pending_jobs), FIX-2 (transpile to ISA before submission), FIX-3 (robust token loader). No physics impact.
- Simulator (`AerSimulator`) confirms the exact extraction path and |M| = 4. Hardware path (SamplerV2, Batch) is identical to prior validated experiments.

7. ProofRecord construction (`bellwether-proofrecord-2.0`)

```
raw_counts_hash      = H( canonical(counts_by_setting) )
calibration_hash     = H( canonical(calibration_snapshot) )
witness_hash         = H( canonical(multipartite_witness) )  # contains M, E_values,
bounds
mermin_nonce         = H( lp(raw_counts_hash) || lp(job_id) || lp(calibration_hash) ||
lp(witness_hash) )
record_hash          = H( canonical(all fields except record_hash) )
```

Tampering with the measured statistic breaks `mermin_nonce` ; any covered-field edit breaks `record_hash` . Verified by 10/10 pre-lock mock tests (tamper-evidence over both counts and witness) and by the independent `verify.py` .

8. Commercial purpose (Covenant requirement)

BELLWETHER-2 extends the ExecutionProof patent family with a **multipartite** entropy/authorization primitive: a ProofRecord whose witness certifies that **three or more independent parties must jointly agree** — cannot be replicated by any pair acting alone. Candidate RF-100 use: **multi-authority / multi-signature authorization** (e.g., 3-of-3 board approval, nuclear two-person rule extended to N parties). Publication is gated pending the file-first IP determination (§9).

9. IP gate

BELLWETHER-2 is **almost certainly new patentable matter**. Per the Covenant it is held **local-only** (no GitHub / Zenodo / LinkedIn / RF-100 disclosure) until Derek makes the **file-first** determination. Execution is **not** disclosure; running the job and fetching results is permitted.

To God be the glory. Proof Before Power. Verification Before Execution.